

Updates and Corrections to the Whitepaper

"Autonomous Microseismic Subsidence Prediction & Early-Warning Network" — reviewed against SIH 2026 PS SIH26025 (Ministry of Coal, Mine Subsidence)

Summary

In its current form, the project is likely to struggle with domain-expert judging because its core sensing premise needs correction: a shallow-buried MEMS IMU (ISM330DHCX class, 70 $\mu\text{g}/\sqrt{\text{Hz}}$ per ST's datasheet) is not well suited to detecting the Mw -2 to -3 microseismic "precursor" events the design centres on. Those events radiate energy at $\sim\text{kHz}$ frequencies, which attenuate by roughly $5\times$ at 100 m and $\sim 3,600\times$ at 500 m through coal-measure overburden, while the MEMS noise floor is roughly $1,000\times$ higher than a geophone's. The design has effectively bet the whole system on the one modality that performs poorly at the surface.

The design also leaves out most of the explicit "Expected Solution" bullets in the problem statement — tilt, inter-node displacement, crack detection, GIS, dashboards, SMS/app alerts, offline+cloud sync, ESP32/Arduino — and SIH is judged in part on adherence to that list by ministry/PSU (Coal India / CMPDI / CIMFR / DGMS) evaluators. The good news: the physically appropriate modality — MEMS tilt/inclination sensing — is straightforward to achieve with the exact IMU already specified, and it is the correct signal for subsidence.

Recommendation: pivot the primary sensing from microseismics to tilt + displacement + crack detection, keep the LoRa mesh / solar-pole / consensus-alert architecture, add the software stack the PS calls for, and correct the cost and Jharia-hazard framing. The $\text{₹}1,200\text{--}1,500/\text{node}$ budget is not achievable as specified, and the Jharia hazard figures cited are accurate but describe coal-fire-driven subsidence rather than the underground-mining-void subsidence the PS targets.

Key Findings

1. The problem statement is identifiable, and the whitepaper answers a slightly different question than the one asked.

The PS is SIH26025, "Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India," submitted by the Ministry of Coal, category Hardware, theme Disaster Management, with an idea-submission window closing 20 September 2026. (The "PS 25" reference maps to SIH26025.) The PS is explicit and prescriptive: it names the sensors (tilt/inclination, vibration, displacement/stretch, crack detection, optional positioning), the comms (LoRa/Zigbee/Wi-Fi mesh), the hardware platforms (Arduino/ESP32/Raspberry Pi), and the full software stack (GIS deformation maps, dashboards for operators/planners/regulators, SMS/email/app alerts, offline capability with cloud sync). The whitepaper delivers essentially one of these (vibration, reframed as microseismics) and omits the rest.

2. SIH judging weights both PS-adherence and novelty — and ministry judges tend to penalise "wrong problem" solutions.

Published SIH evaluation guidance lists criteria including novelty, complexity, clarity in the prescribed format, feasibility, practicability, sustainability, scale of impact, user experience, and future-work potential, scored across multiple rounds (teams scored $\sim 1\text{--}20$ per criterion, weighted to a final score out of 100). Grand Finale judges are drawn from the sponsoring organisation — here that means Coal India Ltd / CMPDI / CIMFR / DGMS domain experts who monitor subsidence professionally. Mentor and finalist accounts consistently stress that juries "ask whether it works and whether you understand the problem," and that a stable working prototype

beats an ambitious non-working one. A headline mechanism that a mining-seismology expert can dismantle in one question carries real risk.

3. The microseismic premise is the main item to correct.

Independent physics check: for a source radius ~ 1 m, corner frequency is $\sim 700\text{--}1,300$ Hz (Brune model), consistent with Gibowicz's (1995, *Pure and Applied Geophysics*) compilation of mining events showing corner frequencies up to $\sim 4,400$ Hz for the smallest events. Using $M_w = (2/3) \cdot \log_{10} M_0 - 6.07$, $M_w -2 \rightarrow M_0 \approx 1.3 \times 10^{11}$ N·m and $M_w -3 \rightarrow M_0 \approx 4.0 \times 10^{10}$ N·m. The unattenuated peak ground velocity of an $M_w -2$ event at 100 m is only $\sim 0.3\text{--}0.6$ mm/s; anelastic attenuation ($Q \approx 100$, $\beta \approx 2,500$ m/s, $f \approx 1,300$ Hz) then cuts the kHz energy by $\sim 5\times$ at 100 m and $\sim 3,600\times$ (≈ 71 dB) at 500 m. The ISM330DHCX noise floor over the bandwidth needed to catch a kHz signal is $\sim 0.7\text{--}4$ mg — its velocity-equivalent floor (~ 10 $\mu\text{m/s}$ narrow-band at 1 kHz) is comparable to or worse than the already-attenuated signal. A geophone is $\sim 1,000\times$ (60 dB) quieter (velocity floor ~ 10 m/s). The literature indicates low-cost MEMS accelerometers detect reliably only down to about ML 1.5–3, and reviews note that <1 ng/ $\sqrt{\text{Hz}}$ floors are needed for weak microseismic motion — roughly five orders of magnitude better than this part offers. Net assessment: surface/shallow MEMS detection of $M_w -2/-3$ events at 100–500 m depth is not physically plausible as designed.

4. The whitepaper contains an internal contradiction worth resolving before submission.

It correctly states that geometric spreading and frequency-dependent attenuation limit reliable precursor detection to "tens of metres, not hundreds" — then separately claims an 80–100 m per-unit coverage radius. These cannot both be true. Either the detection range is tens of metres (so the 100 m coverage claim needs revision and the node count roughly quadruples), or 100 m is the real figure (which would contradict the stated physics). This is worth reconciling before a hostile examiner does it for the team.

5. Real coal-mine microseismic monitoring places sensors in boreholes and underground galleries, not shallow surface soil.

The first Indian longwall microseismic study (Rajendra mine, Shohagpur, 2001–2004) used ten or more uniaxial geophones grouted into 40–60 m boreholes in high-RQD rock. Chinese well-ground systems combine surface arrays with geophones along underground roadways specifically to raise SNR and location accuracy. Borehole placement exists precisely to escape the surface noise and shallow-soil attenuation that the current design's surface siting runs into.

6. The physically well-suited modality — tilt/inclination — was dropped, and it's the easy win.

Per Mine Subsidence Engineering Consultants, maximum subsidence in super-critical extraction is "typically 55% to 65% of the extracted seam thickness" (rarely fully reached because chain pillars provide support), with tilts (change in subsidence over distance, in mm/m) that can reach tens of mm/m — documented cases of 78–95 mm/m in thick, shallow extraction — face advance of $\sim 2.7\text{--}4.8$ m/day (SCCL Adriyala), and troughs developing over days to months. Low-cost MEMS inclinometers resolve $\sim 0.003\text{--}0.01^\circ$ ($\approx 0.05\text{--}0.17$ mm/m). The subsidence tilt signal is therefore 2–3 orders of magnitude larger than the sensor resolution — readily detectable. The same ISM330DHCX already specified works well as a tiltmeter: the right hardware, redirected at the right physics.

7. The Jharia numbers are accurate but should be recharacterised, and the site choice sits awkwardly against the PS scope.

The figures (318 km² analysed; 1.2 km² very-high, 7.3 km² high, 40.2 km² moderate hazard) are accurately quoted from Karanam, Motagh, Garg & Jain (2021), *International Journal of Applied Earth Observation and Geoinformation* — but that paper's hazard zonation is coal-fire-induced subsidence, with the 1.2 km² very-high zone defined as "where both the coal fires and land subsidence hazards are prominent." Jharia's dominant, best-known subsidence driver is its century-old underground coal fires rather than clean mining-void collapse, and Jharia also has extensive opencast mining. The PS is specifically about underground coal mines; the whitepaper's own title reportedly spans "open-pit and underground," widening the mismatch. Using Jharia as the flagship case study invites a predictable question: "Is your system detecting mining subsidence or fire subsidence, and how do you tell them apart?"

8. The cost claim needs revising.

A bottom-up BOM at current Indian pricing: a combined ESP32+SX1276 LoRa board retails around ₹1,850 (Robocraze/IndiaMart); a bare ESP32 (~₹350–450) plus a bare RA-02/SX1278 (~₹200–300) is cheaper but still consumes most of the budget before sensing is added. An industrial ISM330DHCX is ~US\$4.45 (~₹380) as a bare chip at volume (LCSC) but ~US\$20 (~₹1,700) as a usable breakout (Adafruit/SmartElex) — and it still needs a PCB. Add a solar panel (~₹150–300), 18650 Li-ion cell (~₹150–250), TP4056 charge management (~₹30–50), IP-rated enclosure (~₹200–500), 1 m metal pole (~₹200–400), and LEDs/misc (~₹50–100). A realistic per-node cost is roughly ₹2,500–5,000 at prototype/retail scale, and even an optimistic volume build lands well above ₹1,500. Fitting the industrial IMU inside ₹1,200–1,500 alongside the MCU, radio, solar, battery, enclosure and pole is not achievable as currently specified.

9. The node-count geometry is optimistic, and power feasibility needs a closer look.

For gap-free coverage of 48.7 km² with a genuine 100 m radius, hexagonal covering (area per node $\approx 2.598 R^2 \approx 0.026 \text{ km}^2$) needs ~1,870 nodes — more than the stated 1,500–1,700, which corresponds to the naive πR^2 (0.0314 km²/node $\approx 1,550$) estimate that leaves coverage gaps. If the honest detection radius is "tens of metres," the count grows to many thousands. Separately, continuous high-rate IMU sampling plus on-device ML is not a low-power duty cycle — solar LoRa nodes achieve multi-hundred-day autonomy (e.g., ~476 days on a 3,000 mAh cell) precisely by sleeping more than 95% of the time; always-on kHz sampling to catch microseismic events breaks that budget. The "trend-based, relaxed-latency" framing helps the radio duty cycle but not the sensing/compute power draw.

Details

PS alignment, and how the project will actually be judged

SIH runs an internal college hackathon (SPOC-nominated teams), a national screening of submitted ideas, and a Grand Finale where shortlisted teams build and present to expert panels at nodal centres. Evaluation criteria published across editions include novelty, complexity, clarity in the prescribed format, feasibility, practicability, sustainability, scale of impact, user experience, and future-work potential, scored in rounds and weighted to 100. Because the sponsoring body for SIH26025 is the Ministry of Coal, the finale jury will very likely include people from Coal India / CMPDI / CIMFR / DGMS. Two implications follow: (1) the "Expected Solution" bullet list functions as a de facto requirements spec — omitting GIS, dashboards, alerts, offline+cloud sync, and the named modalities can read as not having fully addressed the ask; (2) novel technical approaches are rewarded only if they hold up under expert scrutiny. A novel-but-uncertain core mechanism is a weaker position than a conventional-but-sound one here.

Why surface microseismics is the item to fix (the numbers)

Radiated energies are ~ 60 J ($M_w -2$) and ~ 2 J ($M_w -3$). Corner frequency for a ~ 1 m source $\approx 0.37 \cdot \beta/r \approx 700\text{--}1,300$ Hz. Unattenuated PGV: $M_w -2$ at 100 m $\approx 0.3\text{--}0.6$ mm/s; $M_w -3$ at 500 m $\approx 0.02\text{--}0.04$ mm/s. Attenuation $\exp(-\pi f R/Q\beta)$ gives ~ 0.20 at 100 m and $\sim 2.8 \times 10^{-4}$ at 500 m. The MEMS floor ($70 \mu\text{g}/\sqrt{\text{Hz}}$, confirmed on ST's product page as "low noise ($70 \mu\text{g}/\sqrt{\text{Hz}}$)" with 2 kHz bandwidth) integrated over ~ 1 kHz ≈ 2.2 mg; even narrow-band (~ 100 Hz) ≈ 0.7 mg, velocity-equivalent $\sim 10 \mu\text{m/s}$ at 1 kHz — comparable to or larger than the surviving signal even in the best case ($M_w -2$ at 100 m, $\text{SNR} \approx 1$). This is corroborated by mining-seismology practice: near-source/borehole arrays detect below $M_w -2$ (Underground Research Laboratory, Manitoba; TauTona catalog completeness to $\sim M_w -1.3$), a level that surface soil-buried MEMS cannot approach. Bandwidth is not the limiter (the IMU can sample to a few kHz); the limiters are the noise floor and the kHz attenuation over 100–500 m of overburden plus the shallow soil.

The modalities the PS asks for, and their feasibility

- **Tilt/inclination (the recommended primary channel):** Low-cost MEMS inclinometers achieve $0.003\text{--}0.01^\circ$ resolution (e.g., the LARA multi-IMU device reports 0.003°); subsidence tilts reach mm/m to tens of mm/m. This is the most detectable, cheapest, lowest-power precursor, and it directly reuses the IMU already in the design. Note the well-documented need for thermal compensation of MEMS tilt in outdoor conditions.
- **Inter-node displacement:** RTK GNSS (u-blox ZED-F9P) gives cm-level 3D positioning; a simple RTK2B-class board is $\sim$ US\$150–200 plus a multiband antenna. Too costly for every node, but well suited to a sparse subset of "anchor" nodes measuring absolute subsidence/horizontal strain, with cheaper nodes handling relative tilt. LoRa/UBW ranging is far coarser and won't resolve mm-scale movement.
- **Crack detection:** Strain gauges, wire/potentiometric extensometers, and conductive-trace or resistive/capacitive crack sensors are inexpensive (tens to low-hundreds of rupees) and map directly onto the PS's "crack initiation" requirement and onto the tensile-strain zone at panel edges.
- **Vibration:** Worth keeping a MEMS accelerometer for gross ground-vibration/roof-fall energy and blast discrimination — presented as coarse vibration sensing rather than foreshock microseismics.

Prior art, and how to keep the project distinctive

CIMFR's Mine Subsidence & Surveying division already conducts Indian subsidence monitoring via Total Station and GNSS, and has built 3D prediction models (Modified Influence Function, Non-Effective Width concept) for single/multi-seam Indian conditions. CMPDI/CIMFR/IIT-ISM run active subsidence R&D (e.g., paste-fill panel monitoring at Shyampur B/Mugma). LoRa/IoT wireless sensor networks for landslides and ground movement are also well published (e.g., the L'Aquila LoRa landslide network; Geocube). So a "LoRa + MEMS tilt array + dashboard" is not novel on its own. Genuine novelty levers worth foregrounding: (1) the consensus "council" alerting, repurposed for tilt/strain anomaly voting to suppress false alarms — a real contribution; (2) fusing an in-situ dense tilt/crack array with satellite InSAR (how Jharia is actually monitored today) to give both wide-area context and local early warning; (3) an offline-first, low-cost, student-buildable node genuinely designed around DGMS workflows.

Regulatory and deployment considerations

LoRa in India uses the license-free 865–867 MHz band (WPC/DoT), up to 30 dBm/1 W ERP, license-free for low-power nodes; secondary sources describe the duty-cycle guidance inconsistently (commonly cited as 1%/hour, though some note no explicit statutory duty-cycle limit in the Indian rule, unlike the EU). DGMS approval tends to be conservative: at least one CIL R&D field trial was recorded as stalled because "DGMS not convinced to grant permission for field trial," which is a reminder that regulator buy-in is a real gate — so the regulator-facing interface the PS explicitly asks for is not optional garnish. Jharia's surface is densely populated

with settlements over and around the workings, raising land-ownership, agricultural-land, habitation, theft and vandalism considerations for the many exposed solar poles — a deployment risk worth addressing explicitly (tamper detection, community engagement, minimal-footprint siting).

Recommendations

Stage 1 — Reframe before building (ideally well before the internal hackathon)

- Demote microseismics from "primary sensing element" to an optional coarse-vibration channel. Make tilt/inclination the primary precursor, with crack/strain second and relative/absolute displacement third. This aligns with both the physics and the PS bullet list, and requires no new core hardware — the ISM330DHCX becomes a tiltmeter.
- Rewrite the case study. Either (a) keep Jharia but explicitly separate fire-driven from mining-void subsidence and cite the hazard figures correctly as coal-fire hazard zonation (Karanam et al., 2021), or (b) switch the flagship example to a cleaner underground-longwall case (e.g., SCCL Godavari Valley / Adriyala, or an ECL/BCCL underground panel) that matches the "underground coal mines" scope. Drop "open-pit" from the title.

Stage 2 — Build out the PS-complete stack (currently missing pieces)

- Add GIS deformation/risk-zone maps, a web + mobile dashboard with distinct operator/planner/regulator views, SMS/email/app alerting, and offline logging with periodic cloud sync. These are individually straightforward and make up half of the PS's "Expected Solution."
- Name and use ESP32 (and Raspberry Pi as gateway) explicitly — the PS asks for it and it signals student-buildability.
- Keep the LoRa mesh, solar poles with hazard LEDs, on-device noise filtering, and the consensus "council" alert logic — these are genuinely strong and worth foregrounding.

Stage 3 — Correct the numbers

- Publish an honest BOM at a realistic ~\$2,500–4,000/node (or justify a lower target with a specific volume-sourced chip-down design), and separate one-time vs. recurring costs. Present a tiered architecture: many cheap tilt/crack nodes plus a few RTK-GNSS anchor nodes to control cost.
- Recompute coverage with hexagonal packing and a defensible per-node radius; give power-budget math showing duty-cycled sensing sustains solar operation.

A convincing prototype demo (what to show judges)

- **Bench:** a pole-mounted node detecting sub-0.1° induced tilt, relaying over a 3-node LoRa mesh, with the consensus logic suppressing a single-node false alarm and escalating a genuine multi-node event to a gateway → dashboard → SMS alert.
- **Field/sandbox:** 4–6 poles in a soil bed or slope; induce progressive tilt/settlement and a surface crack; show the live GIS map colouring a risk zone, a rising trend (tilt rate + crack width), and an automated early-warning alert, plus offline buffering with delayed cloud sync when connectivity returns.

Benchmarks that would change this assessment: if the team can produce field data showing a shallow MEMS node reliably resolving sub-Mw-0 events at ≥ 100 m, microseismics could return as primary. If a full node can be sourced under \$1,500 with bench-verified power autonomy, the cost claim stands as written. Absent both, the pivot above is recommended.

Caveats

- Physics estimates are order-of-magnitude. The PGV/attenuation figures swing by factors of several with assumed shear velocity (2,000–3,000 m/s), stress drop, radiation pattern, and especially Q (50–150). The qualitative conclusion — non-detectable at the surface, dominated by kHz attenuation and MEMS noise floor — is robust across the full plausible range; the exact SNR is not.
- Jharia area figures vary by source (analysed area 318 km²; coalfield extent cited variously ~280–450 km²) because different studies analyse different footprints — the team should state which footprint and which study it uses.
- Duty-cycle regulation for India's 865–867 MHz band is described inconsistently across secondary sources (1%/hour vs. no explicit statutory limit); verify against the current WPC notification before making compliance claims.
- PS numbering: "PS 25" corresponds to SIH26025; confirm the exact ID on sih.gov.in before submission.
- This review was based on the user's summary rather than the team's full 17-page whitepaper; some omissions flagged here may be present but under-emphasised in the original document.